The Spaghetti Engine
VISUAL SYSTEM MODELLING WORKSPACE

The Spaghetti Engine

Development Log

Fuzzy Cognitive Map modelling tool

Field	Value
Version	0.5.1 (prototype)
Log entry date	5 November 2026
Status	Released: deployed by the push of this release to the GitHub main branch (the commit titled Release 0.5.1), together with the 0.5.1 documents and this log. Netlify deploys the site automatically.
Live URL	spaghetti-engine.netlify.app (Netlify, auto-deployed from the GitHub main branch)
Source repository	github.com/aPaarl/Spaghetti-engine, main branch (Release 0.5.1)
Previous version	0.5.0, 3 November 2026, commit 57f148c (see Spaghetti_Engine_Dev_Log_v0.5.0.docx)
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1. Purpose of This Log

This Development Log records the changes from version 0.5.0 (3 November 2026) to version 0.5.1 of The Spaghetti Engine. Version 0.5.1 makes the tool usable on tablets and phones, adds a presentation view of the network map for workshops, and replaces the browser's pop-up messages with messages inside the tool.

The log documents what changed and why, how existing results are affected, how the changes were checked, and the deployment status.

2. Summary of Version 0.5.1

Area	Change
Tablets and phones	Compact header, one scrolling row of tab groups, a one-line scope bar, the Inspector below the map, one-finger panning on touch screens, and stacked panels on narrow screens.
Presentation view	A Present button on the Network tab: the map alone, full screen, with larger text, a title and legend, and no editing.
In-app messages	The remaining browser pop-ups (failed import, failed workbook export, analysis errors, the PDF fallback, the recovery screen) are messages inside the tool; a failed PNG export of the map, silent before, is now reported.
Record of 0.5.0	Version 0.5.0 was pushed as commit 57f148c on 3 November 2026.

3. The Changes in Detail

3.1 Tablets and phones

- **Header.** Below 640 pixels the version label and the Guide label are hidden and Export JSON and Import are reached through the Models dialog, which already offered both; below 1024 pixels Export Analysis Workbook, Export JSON, and Import show only their icon; the save time is shown from 1280 pixels.
- **Tabs and scope bar.** The tab groups form one row that scrolls sideways on a phone. Below 1024 pixels the analysis scope bar shows only "Viewing:" and the view's name, with a Scope button that opens the counts, the scope menu, and the module chips.
- **Network tab.** Below 1024 pixels the map card is 78% of the screen height (at least 440 pixels) and the Inspector sits below it instead of beside it. On a touch screen (pointer: coarse) one finger pans the map instead of drawing a selection box, and dragging a concept still moves it. On a phone the toolbar buttons Add concept, Present, and Export PNG show only their icons.
- **Other tabs.** Grids of three or four columns (system statistics, settings, result cards) become one or two columns on a narrow screen; the two-column panels of Baseline Equilibrium and the scenario list of Scenarios &

Simulation stack below 1024 pixels; the squash switch row wraps; the page padding and the footer text are shorter.

3.2 Presentation view

- **What it shows.** The map alone, fixed over the whole window and in the browser's full screen where allowed, fitted to the screen; the model's name, its counts, and a legend of modules and relationship signs. Concept names are 17 pixels instead of 13 (18 instead of 14 in Editing style) and weights 13 instead of 10.5.
- **What it does not allow.** Concepts cannot be dragged, connected, or selected, the connection dots, zoom controls, and overview map are hidden, and keyboard shortcuts are off; dragging pans, scrolling zooms, and hovering still highlights a concept's connections.
- **Leaving.** The Exit button, Escape, or leaving the browser's full screen returns to the Network tab, refitted to the canvas.

3.3 Messages inside the tool

- **Notices.** A notice stack in the top-right corner replaces alert(). An error stays until closed; a note disappears after 7 seconds; at most four are shown. Each message says what failed and that the model is unchanged.
- **Where they are used.** Monte Carlo and Transition Pathways errors, a file that cannot be imported, a failed workbook export, PDF export falling back to PNG in a browser without CompressionStream, and a failed PNG export of the network map. The recovery screen shows its messages on the screen itself.

4. Implementation Choices

Topic	Choice in the tool	Reason
How messages reach the stack	A window event (se-notice) that any part of the code can send.	Errors arise in analysis runners and export helpers outside the React tree; no component has to pass a callback down.
Which devices count as touch	The CSS media query pointer: coarse.	A laptop with a touch screen and a mouse keeps the mouse behavior; a tablet or phone gets one-finger panning.
Export and Import on a phone	Only in the Models dialog.	The header then fits on two rows, and the dialog already lists both actions with their full names.
Full screen	The Fullscreen API where allowed, with the fixed overlay alone as fallback.	Some browsers (and an embedded preview) refuse full screen; the presentation still fills the window.

5. Effect on Existing Results

- No analysis result changes. Saved models, settings, and exports are unchanged.

- On a computer screen of 1024 pixels or wider the layout is as in 0.5.0, apart from the Present button and the header's save time, now shown from 1280 pixels.
- Version raised from 0.5.0 to 0.5.1 in package.json and package-lock.json.

6. Defects Found and Fixed

- **Silent PNG export.** A failed PNG export of the network map gave no feedback; it now shows a message.
- **Presentation fit.** During development the map was fitted before the full-screen stage had its final size; it is now refitted whenever the stage is resized while presenting.
- **Network tab crash during development.** An effect of the presentation view read the React Flow instance before it was defined, which stopped the tab; fixed before release.

7. Verification

- **Tests:** tests/test_marginal_benefit_tp.cjs, tests/test_transition_direction.cjs, tests/test_pathway_blank_inputs.cjs, tests/test_pathway_expert_robustness.cjs, and tests/test_fcm_layout.mjs pass.
- Layout, in the production build with separate browser storage: every tab at 375 pixels (phone) and 768 pixels (tablet) wide, checked for content wider than the screen (none), and at 1024 pixels against 0.5.0. Touch behavior was checked with an emulated touch screen, not on a physical device.
- Presentation view entered with Present and left with Exit and with Escape; the map fills the window and concepts cannot be dragged. The import message was shown for a file that is not a model.
- **Build:** the production build completes without errors.

8. Deployment Status

- Deployed with the push of the release commit to main on 5 November 2026, together with the 0.5.1 documents and this log. Netlify rebuilds the site automatically.
- **Files changed:** src/SpaghettiEngine.jsx, package.json, package-lock.json, and README.md; with the documents, src/docs-manifest.json, public/sitemap.xml, and the three PDFs in public/docs, where the 0.5.1 versions replace those of 0.5.0.
- **Build note:** npm run build fails in the local OneDrive folder because of the ampersand in the folder path; running node node_modules/vite/bin/vite.js build directly works.
- **PDF note:** the PDFs were printed with Chrome (EDGE_PATH set to chrome.exe).

9. Prototype Scope and Known Limitations

- All limitations listed in the 0.4.0 to 0.5.0 logs still apply.
- On a touch screen there is no hover, so a concept's connections are not highlighted before it is selected; building large models is slower than with a mouse.

- The User Manual and the Methodological Annex still do not describe the expert-defined sequences and the three robustness families of 0.4.4.

10. Related Documentation

Document	Covers
The Spaghetti Engine - User Manual (version 0.5.1)	How to use every feature. Section 1.2 lists the changes of 0.5.1; Section 3.3 the messages; Section 3.8 tablets and phones; Section 4.1 the presentation view.
The Spaghetti Engine - Methodological Annex (version 0.5.1)	The methods; unchanged apart from the version.
Spaghetti_Engine_Dev_Log_v0.5.0	The 0.5.0 release: model library, Getting started guide, Basic and Advanced.
This Development Log	What changed in 0.5.1 and the technical work behind it.

11. Next Steps

- Try the tablet layout and touch editing on a physical tablet in a workshop setting.
- Offer the former peat meadow example as a downloadable file, if wanted, so readers of the case study can open it.
- Describe the expert-defined sequences and the robustness families of 0.4.4 in the manual and the annex.
- Continue with the roadmap: version 0.6.0 (modelling made easier).